

SERVICE MANUAL

Document 5273183

Revision 03

**GE Signa[®] 1.5T
PREMIER 9000
8-CHANNEL CTL COIL**

GE Catalog Part Number: M3335TA

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Rev. 3 November 12, 2009

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警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. - Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatt sérülését eredményezheti.
AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fánleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告
(JA)

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- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
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경고
(KO)

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- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS
(LV)

Šī apkopes rokasgrāmata ir pieejama tikai anglu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

[SPĖJIMAS
(LT)

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL
(NO)

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE
(PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT
(TR)

Bu servis kılavuzunun sadece İngilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu İngilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the 8-Channel CTL Coil, refer to the coil label (see *Figure 1*).

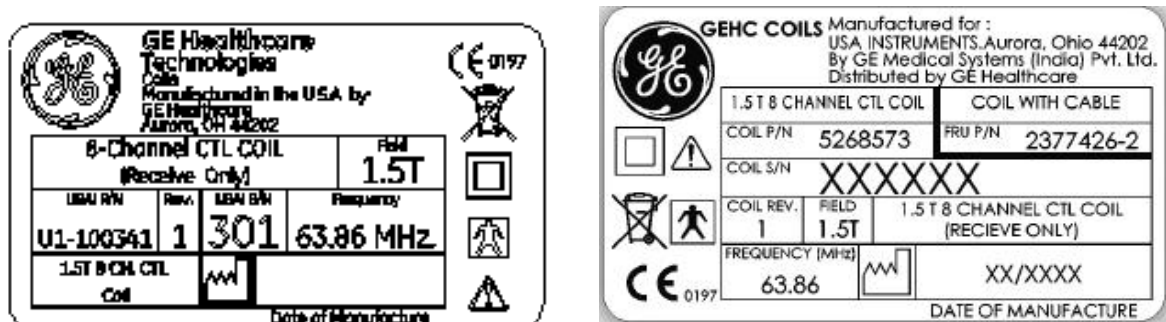


Figure 1: Coil label.

Figure 2 shows a picture of the 8-Channel CTL Coil.



Figure 2: The 8-Channel CTL Coil.

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	GE India Part #	Qty
1	Coil with Coil ID Cable	2377426-2	5268573	1
2	Patient Comfort Pad Kit	2377426-6	5262868	1
1	Adapter Block	2225545-9	5265022	1
1	Operator's Guide	2422287	5267681-299	1
1	Service Manual	2377426-5	5273183	1

Note: Coils do not ship with phantoms. Phantoms come in a unified phantom set with the MR System.

1-2 Compatibility

The 8-Channel CTL coil is compatible with the Excite 1.5T system and requires 10.0 M4 or greater software. 11.0 M4 software is recommended.

The automated QA Tool is only available for software 11.0 M3 and greater.

1-3 Related Documentation

Operator's Guide, GE India Part Number 5267681-299.

Signa Excite Service Methods CD, GE Part Number 2333500-3.

1-4 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room.

Dimensions

43.3" L x 16.1" W x 13.6" H (110.0 cm x 41.0 cm x 34.5 cm)

Weight

18 lbs. (8.1 kg)

1-5 Theory of Operation

The block diagram of the Premier 9000 8-Channel CTL Coil is shown in *Figure 3*. *Figure 4* shows the output diagram, the coil to connector interface is shown in *Figure 5* and *Figure 6* shows the block diagram of the circuit layout. The coil is designed for obtaining diagnostic images of the whole human spine, specifically: cervical, thoracic and lumbar. The coil consists of a main coil base for imaging the entire spine and a removable anterior neck. The coil is a receive-only, phased array 12 element, 8-channel RF coil design. The 12 elements are arranged primarily as a saddle and loop coil configuration. Depending on the anatomy imaged, the user is able to select different imaging modes to selectively image the spine anatomy. The modes are enabled by the clinical user at the console.

The 8-Channel CTL Coil is actively decoupled from the RF transmit coil during transmit by means of a RF choking circuit. The choking circuit elements are switched on and off by pin diodes or by small-signal diodes. During body coil transmission, the pin diodes are turned on, as shorted circuits, with a forward bias DC source provided by the MR system (active decoupling). The small-signal diodes are turned on by the induced RF voltage in the chokes, coupled from the transmit field (passive decoupling). When either the pin diodes or the small-signal diodes are turned on, each of these RF choking elements becomes very high in impedance (typically above 2 kilo-ohm), compared with the other circuit elements (less than 70 ohm). These high impedance elements in the coil segregate the coil circuitry into several isolated electrical segments, preventing any current flow in the coil circuit.

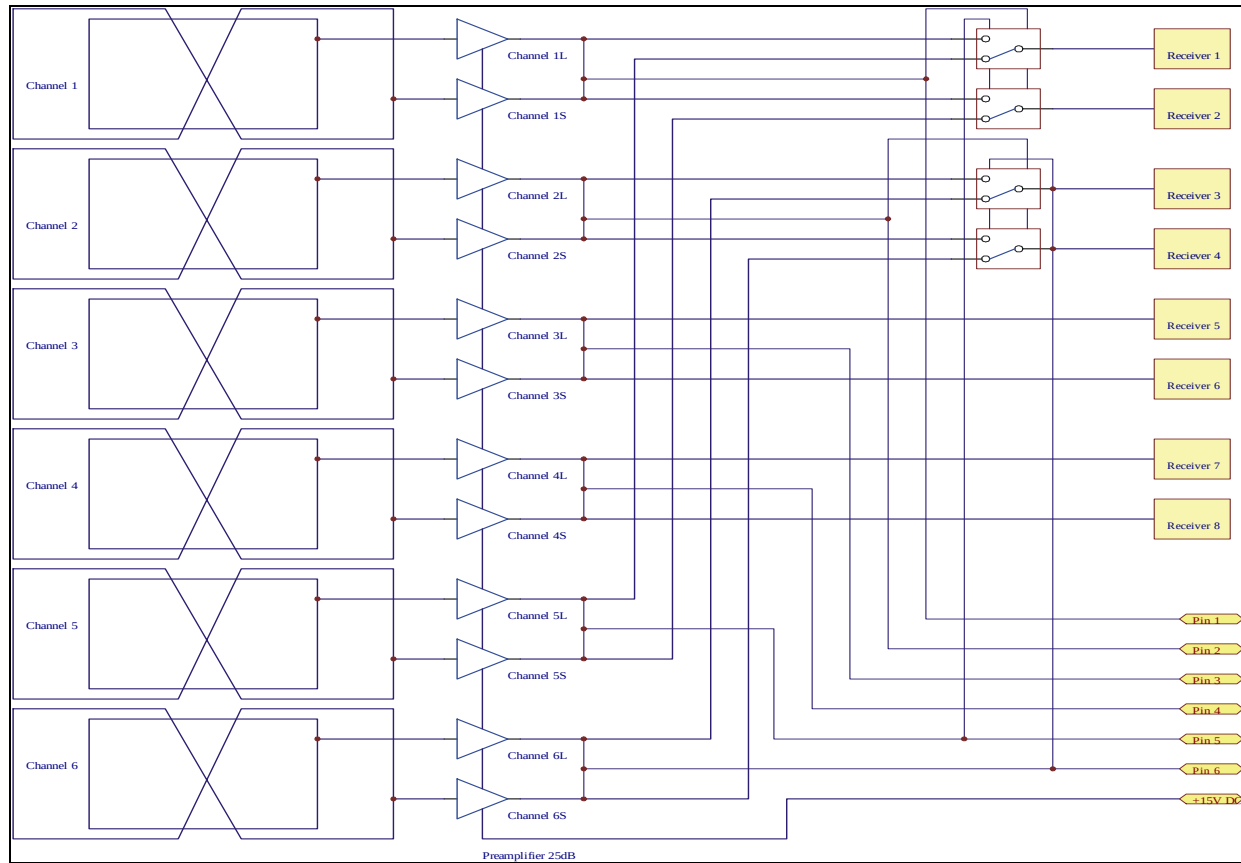


Figure 3: Block diagram of the coil.

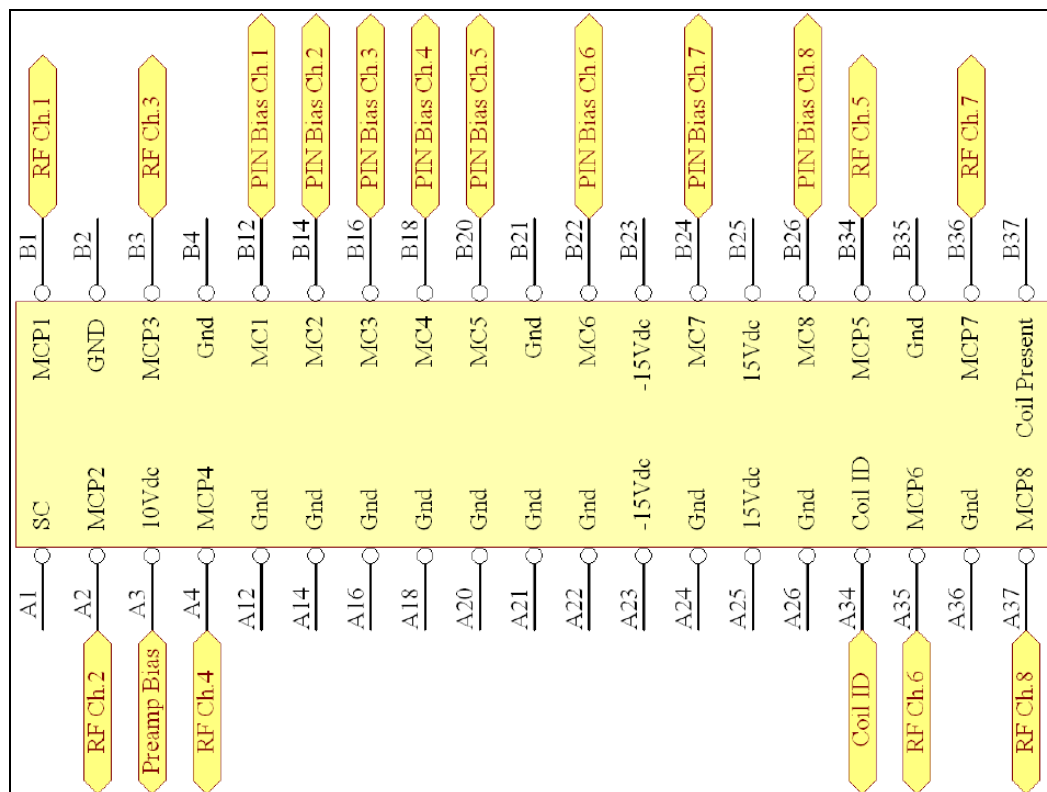


Figure 4: Output diagram.

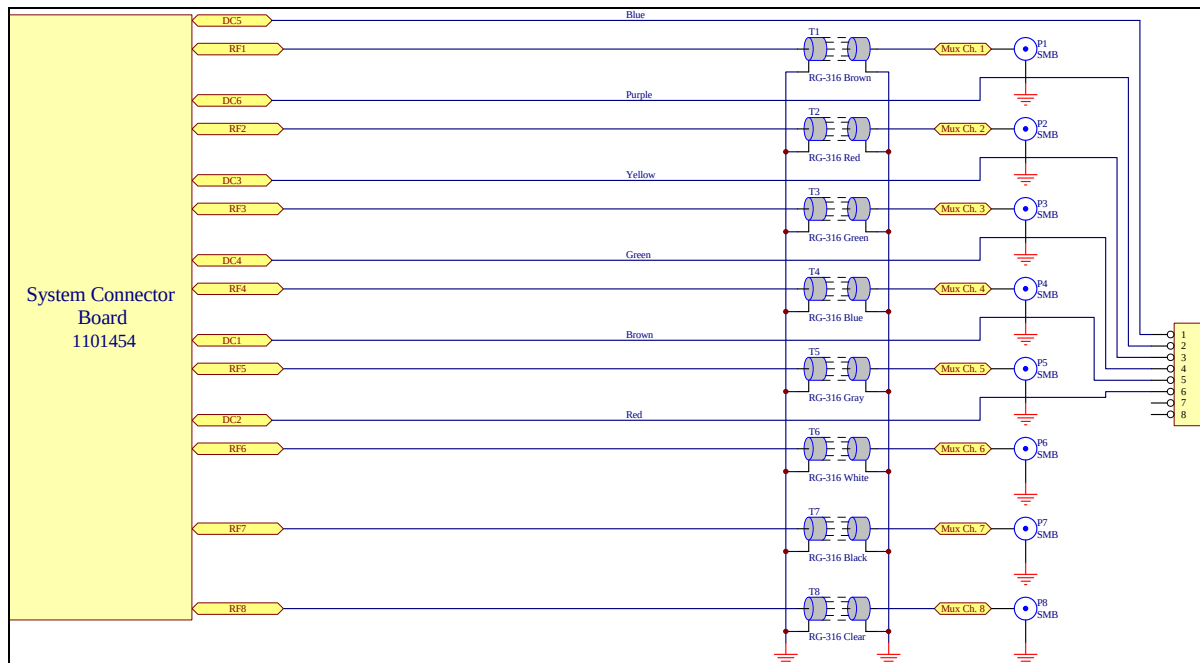


Figure 5: Block diagram of the coil to connector interface.

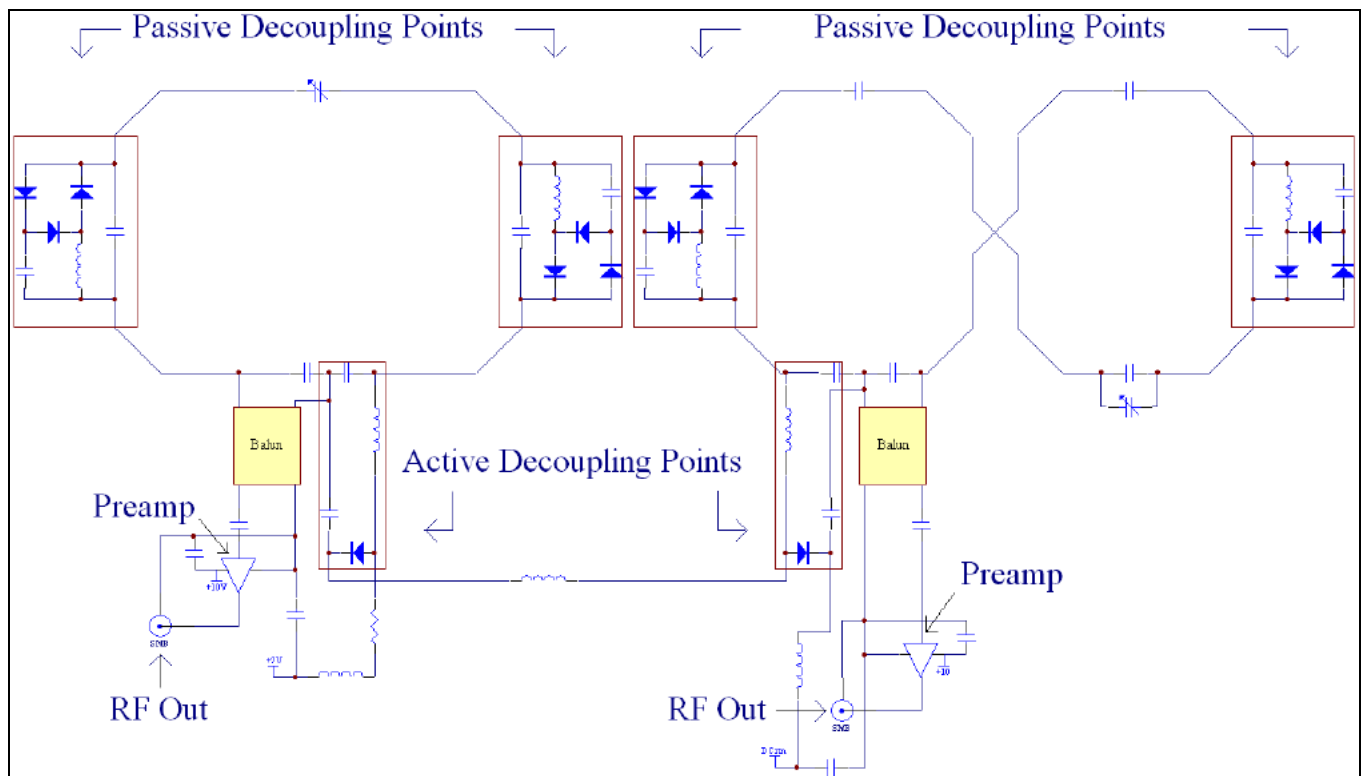


Figure 6: Block diagram of the circuit layout.

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration for 11x Software

The system will automatically recognize the coil using the Coil ID feature. No configuration should be required.

2-1-3 Configuration for 10x Software

The 8-Channel CTL coil requires 10.0 M4 or greater software. 11.0 M4 software is recommended.

Updated coil configurations in the system are required for software versions prior to 11.0 M4. Updated coil configurations are recommended for 11.0 M4 and greater software revisions.

To update the coil configuration files:

1. Using the configuration manager, remove all currently installed (if any) configurations for the 8-Channel CTL coil.
2. Update the coil configurations to the latest version (Refer to Section 9 of “Host Software Install Release 10.X”, direction OW2SCA8G, as found on the Service Methods CD, GE Part Number 2333500-3).
3. Install the new configurations listed in Table 2-1-3.

CONFIGURATION MODES – TABLE 2-1-3

8CTLTOP	8CTLMID	8CTLBOT	8CTL123	8CTL234	8CTL345	8CTL456	8CTL12	8CTL23	8CTL34	8CTL45	8CTL56
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2-2 Installation Functional Checks

1. From the Scan Desktop, start new scan by selecting **[New Pt]**; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click **[Patient Position]** to open protocols window.
2. Plug the coil into the patient table.
3. At the console, verify the Coil has been properly identified by the system: correct name in **Coil** field and correct picture on the screen.
4. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED USING LEGACY PHANTOM – TABLE 3-2-1-1

Description	GE Part #	GE India Part #	Qty
Phantom Kit	2422887	5267488	1

TOOLS REQUIRED USING UNIFIED PHANTOM – TABLE 3-2-1-2

Description	GE Part #	Qty
TL Unified Phantom	5343347	2
Large Cylindrical unified phantom	5342679	1

3-2-2 Explanation of Procedure

The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only.

3-2-3 SNR Setup for 11.0 M3A and Greater Software Versions with Legacy Phantom

1. Remove any other surface coils (if present) from the cradle.
2. Place the coil on the patient table. Position coil so that the top table locators on both sides of the coil fit into the second notch from the magnet. Position the phantom set on the coil without the pad. (see *Figure 7*).



Figure 7: Positioning of the coil and the phantoms on the coil.

3. Landmark the coil at the indicated line on the C-Spine Bridge and advance to scan.

3-2-4 SNR Setup for 11.0 M3A and Greater Software Versions with Unified Phantom Set

Visual	Description
 Figure 0-1	1. Place the coil on the patient table and latch it to the table. 2. Place the TL unified phantoms on the centre of the TL section of the coil as shown in the Figure 0-1.



Figure 0-2



Figure 0-3

3. Align the phantoms to the centre in S/I and R/L directions as shown in Figure 0-2 and Figure 0-3.



Figure 0-4

4. Place the large cylindrical unified phantom on the CT section and fix the C-Spine Bridge to the hooks provided as shown in Figure 0-4.

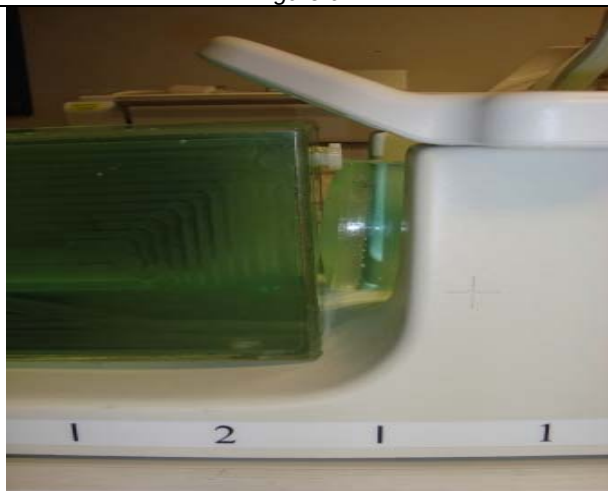


Figure 0-5

5. Adjust the large cylindrical unified phantom such that it is in complete contact with the TL unified phantom (Figure 0-5).
6. Adjust TL unified phantoms such that both the phantoms are in complete contact with each other as shown in Figure 0-6.
7. Connect the coil to respective port of the system LPCA.

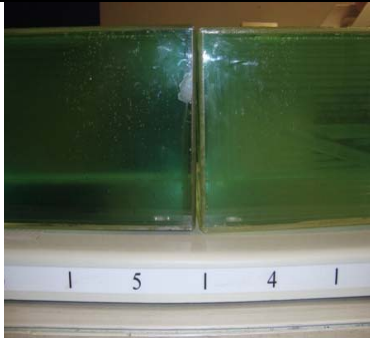


Figure 0-6



Figure 0-7

8. Landmark the coil at the cross mark line on the C-Spine Bridge (Figure 0-7) and advance to scan.

3-2-5 MCQA / SNR Test for 11.0 M3A and Greater Software Versions

1. Move back to the operator workspace and end the previous exam, if required.
2. For 11x/12x systems, go to Common Service Desktop, select **Troubleshooting, Image Quality Troubleshooting Tools**, and then coil **Surface Coils SNR Test**. Continue with Step 3 of this section.

For 14x or later systems, go to Common Service Desktop, select **Image Quality**, and then select **MCQA**. The remaining steps/screen shots do not apply.

If more information is needed for MCQA, it can be found on 14x/15x service CD or on web at: Troubleshooting/Image Quality/multicoil Quality Assurance.
3. When the Multicoil SNR Tool Control window opens, select the coil type you want to run, enter the serial number and select Start Scan. Note: Additional documentation can be accessed by selecting **File, Help, [coil type]**.
4. Follow the reference example below as a visual aid.



- A. From the Common Service Desktop Manager, select the **Image Quality Tab**.

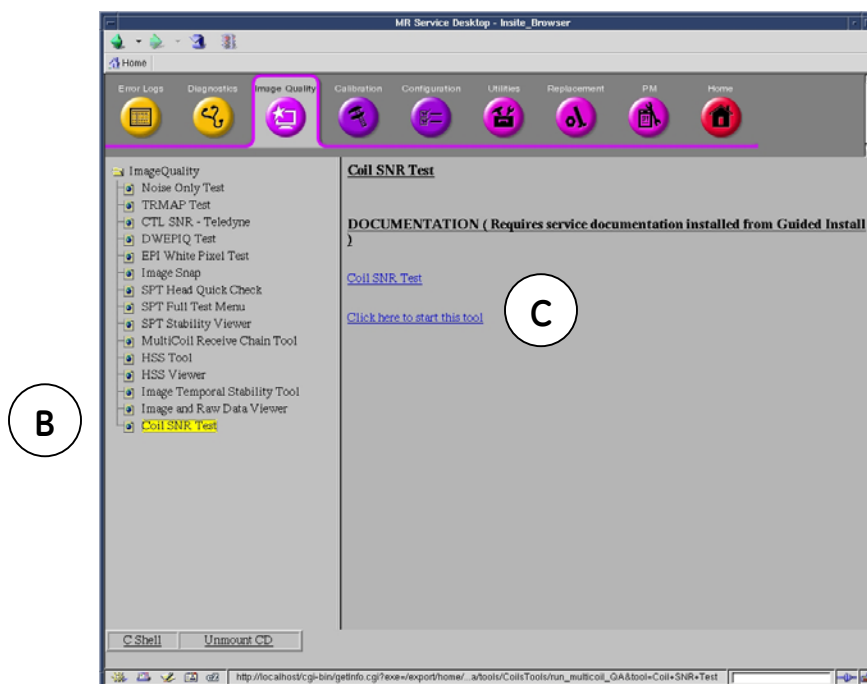


Figure 3.2.5.1

- B. Select **Coil SNR Test**.
- C. Select **Click here to start this tool**. The Multicoil SNR Tool Control displays.

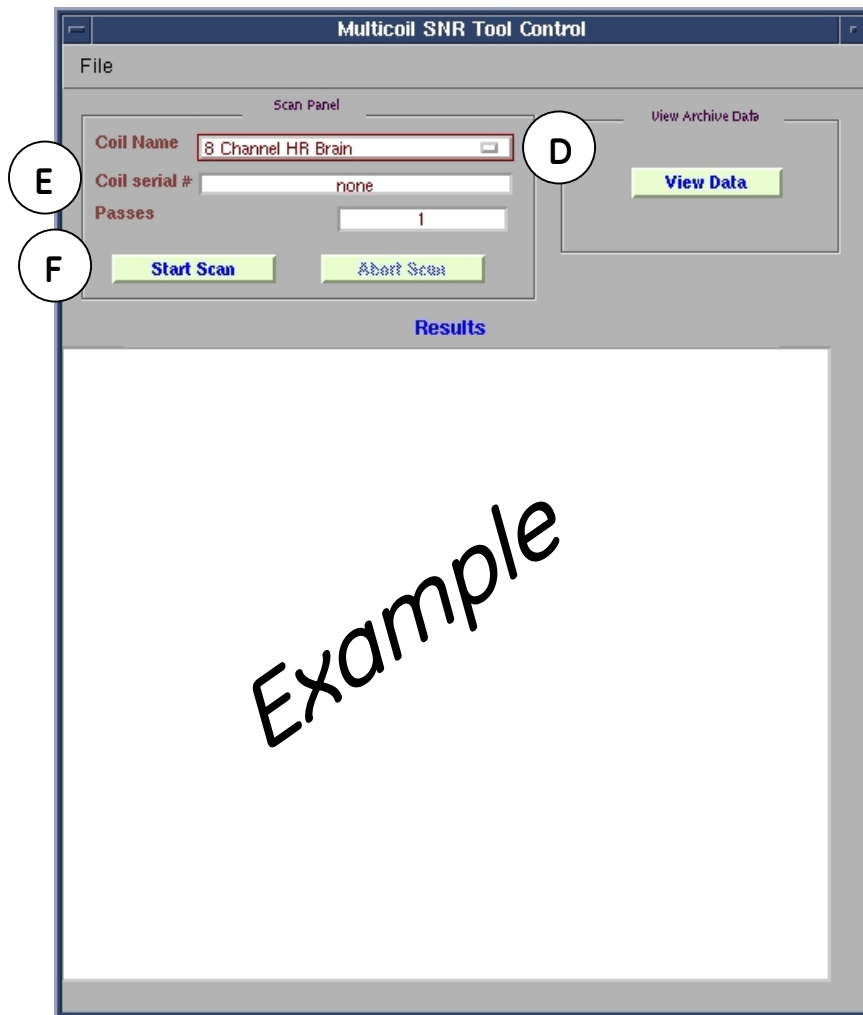


Figure 3.2.5.2: Multicoil SNR Tool Control

- D. The name of the coil should populate the **Coil Name** field. If desired, select the coil you are testing from the dropdown list.
- E. Enter the **Coil Serial #** found on the coil.
- F. Click **Start Scan** to begin the QA scan.

Note: The scan may take several minutes to complete. For example, an 8-channel coil may take approximately 8 to 10 minutes to complete.

- G. The results of the scan display in the Results window with a Passed or Failed indication (refer to Figure 3.2.5.3).

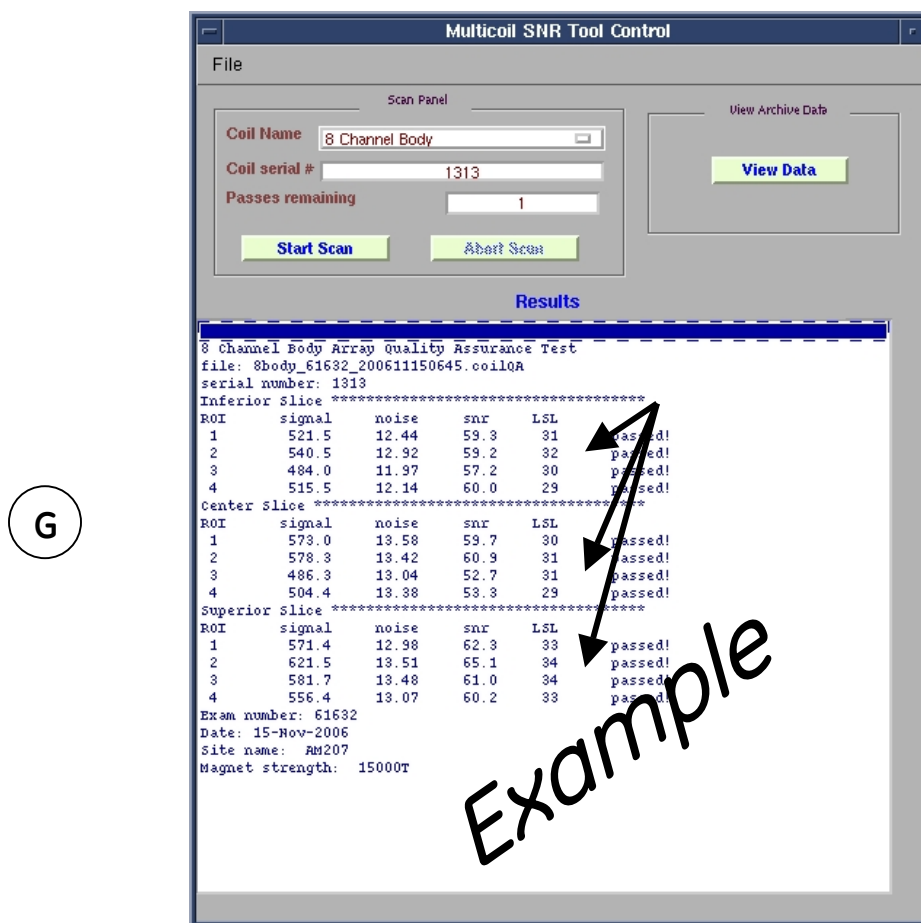


Figure 3.2.5.3: Multicoil Results (Example)

3-3 External Cable Check

Diode Check from the Connector:

Step 1 – Using a multi-meter set to the diode check function, check the coil diodes in the following manner.

Step 2 – Orient the connector as shown in *Figure 8*.

Step 3 – Follow Table 3-3 for the correct pins and lead locations to verify the function of the diodes.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3

PIN Number	Positive Lead	Negative Lead	DMM Reading
1	B12	A12	0.95V +/- 0.2V
2	B14	A14	0.95V +/- 0.2V
3	B16	A16	0.95V +/- 0.2V
4	B18	A18	0.95V +/- 0.2V
5	B20	A20	0.95V +/- 0.2V
6	B22	A22	0.95V +/- 0.2V

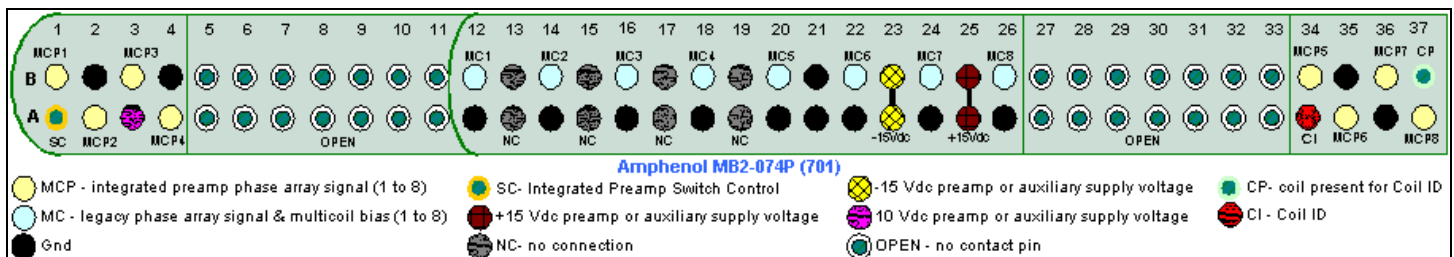


Figure 8: 60 pin Bendix connector.

Note: This connector is system side. Coil side has rows A and B reversed.

3-4 PIN Diodes Check

Same as External Cable Check.

3-5 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again. Make sure the indicator light on the patient table turns green with the proper coil selected on the console.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	If a short or open is observed, check the output cable by itself (see below). If the cable and PIN diodes are okay, then replace the coil.
The output cable has a short or an open.	Disconnect the cable at the coil and check each coaxial line with a DVM (resistance). Try moving and twisting the cable as you watch the meter to find any intermittent connections. Inspect the center pins of the SMB and Bendix connections for wear and proper engagement into the mating connection.	Replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 2-channel coil to see if the same problem occurs.	Correct MRI system problem.
There is a problem with the coil ID; <i>"The prescribed RF coil and detected RF coil do not match."</i>	Check online instructions by accessing the service browser at Diagnostics; Peripherals; SRI Functional Tests; Coil ID	Start the diagnostic by selecting SRI Coil ID and Run . The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the troubleshooting instructions if the correct ID is not displayed. When the troubleshooting is complete and the ID is still not displayed, replace the cable.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise std. dev. measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the holder. Try running the scan without the phantom holder to see if the ghosting diminishes. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

3-6 Coil Element Information

Mode	Rcv 1	Rcv 2	Rcv 3	Rcv 4	Rcv 5	Rcv 6	Rcv 7	Rcv 8	PIN 1	PIN 2	PIN 3	PIN 4	PIN 5	PIN 6	PIN 7	PIN 8
8CTLTOP	ON	ON	ON	ON	ON	ON	ON	ON	-5	-5	-5	-5	+7	+7		
8CTL MID	ON	ON	ON	ON	ON	ON	ON	ON	+7	-5	-5	-5	-5	+7		
8CTLBOT	ON	ON	ON	ON	ON	ON	ON	ON	+7	+7	-5	-5	-5	-5		
8CTL123	ON	ON	ON	ON	ON	ON			-5	-5	-5	+7	+7	+7		
8CTL234			ON	ON	ON	ON	ON	ON	+7	-5	-5	-5	+7	+7		
8CTL345	ON	ON			ON	ON	ON	ON	+7	+7	-5	-5	-5	+7		
8CTL456	ON	ON	ON	ON			ON	ON	+7	+7	+7	-5	-5	-5		
8CTL12	ON	ON	ON	ON					-5	-5	+7	+7	+7	+7		
8CTL23			ON	ON	ON	ON			+7	-5	-5	+7	+7	+7		
8CTL34					ON	ON	ON	ON	+7	+7	-5	-5	+7	+7		
8CTL45	ON	ON					ON	ON	+7	+7	+7	-5	-5	+7		
8CTL56	ON	ON	ON	ON					+7	+7	+7	+7	-5	-5		

SECTION 4 – MAINTENANCE

4-1 Coil Care



Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electrical shock may result if the coil is attached to the system during cleaning or when it is wet.



Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

Step 1 – Remove coil cover.

Step 2 – Remove the two screws that hold down the cable assembly.

Step 3 – Disconnect SMB plugs and the DC Plug on the cable end from the interface boards.

Step 4 – Remove the old cable assembly.

Step 5 – Fix a new cable assembly on the CTL coil and reattach the cable lock with two screws. Connect all eight SMB plugs and the DC Plug to the corresponding receptacles on the interface board (see *Figure 10*).

Step 6 – Close the coil cover.

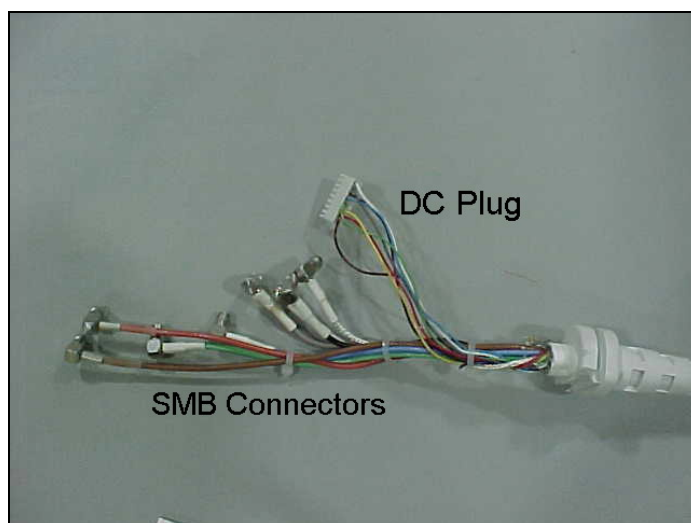


Figure 9: External cable assembly.

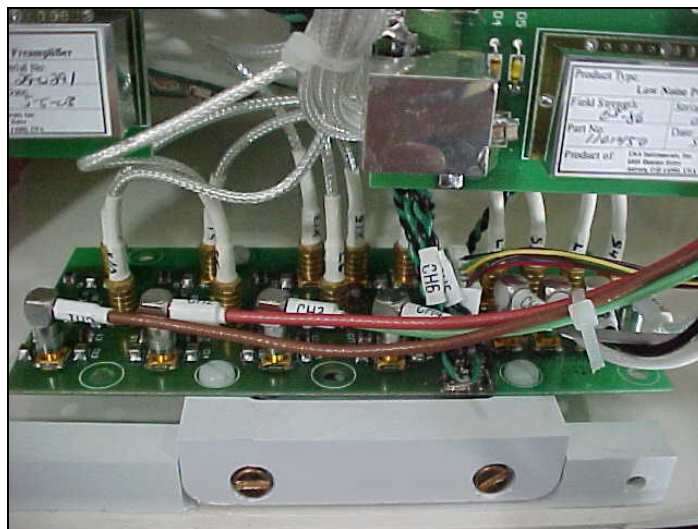


Figure 10: Fixing the cable assembly on the interface board.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #
Coil with Coil ID Cable	2377426-2
Cable Assembly	2377426-7
Adapter Block	2225545-9
Anterior Bridge	2418267
Phantom Kit	2377426-3
TL Unified Phantom	5343347
Large Cylindrical Unified Phantom	5342679
Table Locator Kit	2415553

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Part Name	GE Part #
Patient Comfort Pad Kit	2377426-6

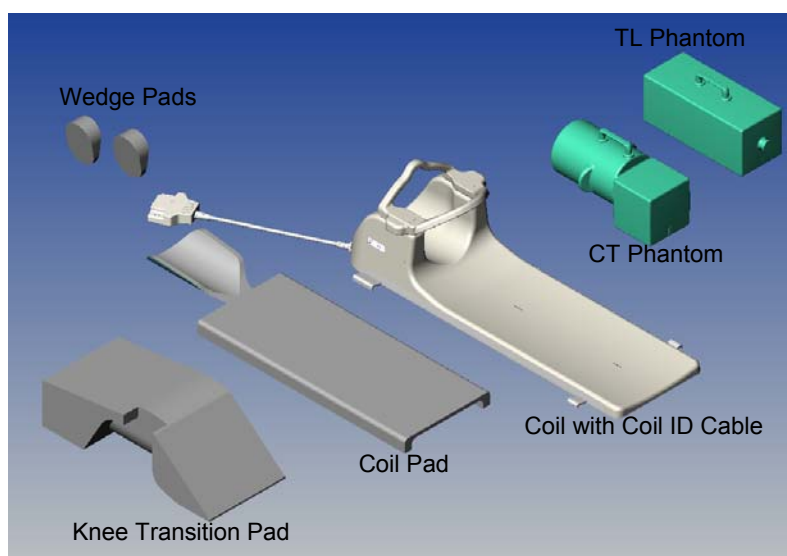


Figure 11: Parts of the coil.

SECTION 7 – APPENDIX**7-1 SNR Data Sheet**

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Date	Configuration	Element/ Roi	Signal Mean	Noise Std. Dev.	SNR	Limit/LSL	Pass/Fail
	8CTL12	1					
	8CTL12	2					
	8CTL34	3					
	8CTL34	4					
	8CTL56	5					
	8CTL56	6					

Comments	
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Exam No. _____

Site Name: _____

Date: _____

Magnet Strength: _____

Date	Configuration	Element/ Roi	Signal Mean	Noise Std. Dev.	SNR	Limit/LSL	Pass/Fail
	8CTL12	1					
	8CTL12	2					
	8CTL34	3					
	8CTL34	4					
	8CTL56	5					
	8CTL56	6					

Comments	
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Exam No. _____

Site Name: _____

Date: _____

Magnet Strength: _____

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	02/08	Tricia Luthringer	First Issue
2	05/08	Tricia Luthringer	Part number changes: Table 1-1, Section 1-3, Section 3-2-1.
3	11/09	Rebecca David	Updated Damage in Transportation, Language Warning, updated shipping list, inserted unified phantom set-up for MCQA per the Phantom Elimination Program